



**Kuwait University**  
**Faculty of Science**  
**Department of Mathematics**

**MATH111: Linear Algebra, Semester: Summer 2022**  
**Final Exam: Sunday, August 13, 2022, 2pm-4pm. Du-**  
**ration: 120 minutes.**  
**Solution key.**

|             |  |  |  |  |  |            |  |  |  |  |
|-------------|--|--|--|--|--|------------|--|--|--|--|
| Name        |  |  |  |  |  |            |  |  |  |  |
| Student ID  |  |  |  |  |  |            |  |  |  |  |
| Section No. |  |  |  |  |  | Serial No. |  |  |  |  |

- Give full justification for your answers and state clearly any theorems you use.
- Calculators and communication devices are not allowed in the examination.

| Problem | Points | Score |
|---------|--------|-------|
| 1       | 30     |       |
| 2       | 15     |       |
| 3       | 15     |       |
| 4       | 20     |       |
| 5       | 10     |       |
| 6       | 10     |       |
| Total   | 100    |       |

**Problem 1**

[30 pts]

$$\text{Let } A = \begin{bmatrix} 8 & 0 & 9 \\ 1 & 3 & 1 \\ -6 & 0 & -7 \end{bmatrix}.$$

(a) [10 pts] Find the eigenvalues of  $A$ .

$$\begin{aligned} 0 &= \begin{vmatrix} 8-\lambda & 0 & 9 \\ 1 & 3-\lambda & 1 \\ -6 & 0 & -7-\lambda \end{vmatrix} = (3-\lambda) \begin{vmatrix} 8-\lambda & 9 \\ -6 & -7 \end{vmatrix} \\ &= (3-\lambda) [(8-\lambda)(-7-\lambda) - (-6)9] = (3-\lambda) [-56 - 8\lambda + 7\lambda + \lambda^2 + 54] \\ &= (3-\lambda) [\lambda^2 - \lambda - 2] = (3-\lambda)(\lambda-2)(\lambda+1). \\ \text{So, } \lambda_1 &= 3, \lambda_2 = 2, \lambda_3 = -1. \end{aligned}$$

(b) [10 pts] Find the eigenvectors of  $A$ .

$$\begin{aligned} \text{Solving } (A - \lambda_i I)\mathbf{x}_i &= \mathbf{0} \text{ for } i = 1, 2, 3 \text{ yields } \mathbf{x}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} \frac{-3}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix}, \\ \mathbf{x}_3 &= \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}. \end{aligned}$$

(c) [4 pts] Find a matrix  $P$  and a diagonal matrix  $D$  such that  $P^{-1}AP = D$ .

$$P = \begin{bmatrix} 0 & \frac{-3}{2} & -1 \\ 1 & \frac{1}{2} & 0 \\ 0 & 1 & 1 \end{bmatrix} \text{ and } D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

(d) [3 pts] Compute  $PD$ .

$$PD = \begin{bmatrix} 0 & -3 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & -1 \end{bmatrix}$$

(e) [3 pts] Compute  $AP$  and check if  $AP = PD$ .

$$AP = \begin{bmatrix} 8 & 0 & 9 \\ 1 & 3 & 1 \\ -6 & 0 & -7 \end{bmatrix} \begin{bmatrix} 0 & \frac{-3}{2} & -1 \\ 1 & \frac{1}{2} & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -3 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & -1 \end{bmatrix}$$

Yes, they are equal.

**Problem 2**

[15 pts]

Let  $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ ,  $\mathbf{y} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$ ,  $\mathbf{z} = \begin{bmatrix} -1 \\ 6 \\ 0 \end{bmatrix}$ .

- (i) [5 pts] Are  $\mathbf{x}$ ,  $\mathbf{y}$ ,  $\mathbf{z}$  linearly independent or linearly dependent? Show your work.

RREF:  $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 2 & -1 \\ -1 & 6 & 0 \end{bmatrix} \begin{bmatrix} 1 & -2 & 3 \\ 0 & 2 & -1 \\ 0 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 5 \end{bmatrix}$  Number of leading is equal to 3. This implies linearly independent.

- (ii) [5 pts] Compute  $\mathbf{u} \times \mathbf{v}$  where  $\mathbf{u} = \mathbf{y} - \mathbf{x}$  and  $\mathbf{v} = \mathbf{z} - \mathbf{x}$ .

$$\mathbf{u} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix} - \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \\ -4 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} -1 \\ 6 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 8 \\ -3 \end{bmatrix}.$$

Therefore,  $\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 4 & -4 \\ -2 & 8 & -3 \end{vmatrix} = \vec{i} \begin{vmatrix} 4 & -4 \\ 8 & -3 \end{vmatrix} - \vec{j} \begin{vmatrix} -1 & -4 \\ -2 & -3 \end{vmatrix} + \vec{k} \begin{vmatrix} -1 & 4 \\ -2 & 8 \end{vmatrix} = 20\vec{i} + 5\vec{j} = \begin{bmatrix} 20 \\ 5 \\ 0 \end{bmatrix}.$

- (iii) [5 pts] Compute the area of the triangle  $\mathbf{x}$ ,  $\mathbf{y}$ ,  $\mathbf{z}$ .

$$\text{Area} = \frac{1}{2} \|\mathbf{u} \times \mathbf{v}\| = \frac{1}{2} \left\| \begin{bmatrix} 20 \\ 5 \\ 0 \end{bmatrix} \right\| = \frac{5}{2} \left\| \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} \right\| = \frac{5}{2} \sqrt{17}.$$

**Problem 3**

[15 pts] Let  $A$  be an  $n \times n$  matrix with eigenvalue  $\lambda$ . Let  $k, \mu \in \mathbb{R}$ . Let  $I_n$  be the  $n \times n$  identity matrix.

- (a) [5 pts] Show that  $kA$  has eigenvalue  $k\lambda$ .

Let  $\mathbf{x}$  with  $A\mathbf{x} = \lambda\mathbf{x}$ . Thus  $kA\mathbf{x} = k\lambda\mathbf{x}$ , so  $(kA)\mathbf{x} = (k\lambda)\mathbf{x}$ , so  $\mathbf{x}$  is eigenvector of  $kA$  associated with the eigenvalue  $k\lambda$ .

(b) [5 pts] Show that  $A + \mu I_n$  has eigenvalue  $\lambda + \mu$ .

Let  $\mathbf{x}$  with  $A\mathbf{x} = \lambda\mathbf{x}$ . Thus  $(A + \mu I)\mathbf{x} = A\mathbf{x} + \mu I\mathbf{x} = \lambda\mathbf{x} + \mu\mathbf{x} = (\lambda + \mu)\mathbf{x}$ , so  $\mathbf{x}$  is eigenvector of  $A + \mu I$  associated with the eigenvalue  $\lambda + \mu$ .

(c) [5 pts] Let  $B$  be a  $3 \times 3$  matrix with eigenvalues  $\frac{1}{2}$ ,  $-1$ ,  $-2$ . What are the eigenvalues of

$$2B^{-2} - 5I_3?$$

| $\lambda$     | $\lambda^{-2}$ | $2\lambda^{-2}$ | $2\lambda^{-2} - 5$ |
|---------------|----------------|-----------------|---------------------|
| $\frac{1}{2}$ | 4              | 8               | 3                   |
| $-1$          | 1              | 2               | $-3$                |
| $-2$          | $\frac{1}{4}$  | $\frac{1}{2}$   | $-\frac{9}{2}$      |

So, the eigenvalues are  $\{3, -3, -\frac{9}{2}\}$ .

## Problem 4

[20 pts]

Let

$$U = \{(x_1, x_2, x_3, x_4, x_5, x_6) \in \mathbb{R}^6 \mid x_1 = -x_6, x_2 = -x_5, x_3 = -x_4\}$$

and

$$W = \{(x_1, x_2, x_3, x_4, x_5, x_6) \in \mathbb{R}^6 \mid x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 0\}$$

be two subspaces of  $\mathbb{R}^6$ .

(a) [5 pts] Is  $U$  a subspace of  $W$ ? Explain!

Let  $\mathbf{x} \in U$  arbitrary. Thus  $x_6 = -x_1$ ,  $x_5 = -x_2$ ,  $x_4 = -x_3$ . Thus  $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6) = (x_1, x_2, x_3, -x_3, -x_2, -x_1)$ . This implies that  $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = x_1 + x_2 + x_3 - x_3 - x_2 - x_1 = 0$ , and therefore  $\mathbf{x} \in W$ , so  $U \subseteq W$  (this means  $U$  is a subspace of  $W$ )

(b) [5 pts] Is  $W$  a subspace of  $U$ ? Explain!

No: the vector  $\mathbf{x} = (1, -1, 0, 0, 0, 0)$  is in  $W$  (because the coordinates sum up to zero) but not in  $U$  (for instance  $1 = x_1 \neq -x_6 = 0$ ).

(c) [5 pts] What is the dimension of  $U$ ? Find a basis for  $U$ .

Let

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \end{bmatrix}$$

Then  $U = \{\mathbf{x} \in \mathbb{R}^6 \mid A\mathbf{x} = \mathbf{0}\} = \text{nullspace}(A)$ . But the matrix  $A$  is in RREF, and has 3 leading coefficients. Thus,  $\text{rk}A = 3$ . By the rank nullity theorem,  $\text{nullity}(A) = 6 - 3 = 3$ . This means  $\dim(U) = 3$ . A basis for  $U$  is given by the three row vectors of  $A$ .

(d) [5 pts] What is the dimension of  $W$ ? Find a basis for  $W$ .

Let

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

Then  $W = \{\mathbf{x} \in \mathbb{R}^6 \mid B\mathbf{x} = \mathbf{0}\} = \text{nullspace}(B)$ . We solve the system. The system is already in RREF. We have 5 free variables. We set  $x_6 = a$ ,  $x_5 = b$ ,  $x_4 = c$ ,  $x_3 = d$ ,  $x_2 = e$ , and find  $x_1 = -x_2 - x_3 - x_4 - x_5 - x_6 = -a - b - c - d - e$ . Thus

$$W = \left\{ \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -a - b - c - d - e \\ e \\ d \\ c \\ b \\ a \end{bmatrix} \mid a, b, c, d, e \in \mathbb{R} \right\}$$

$$= \left\{ a \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + e \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \mid \begin{matrix} a, b, c, \\ d, e \end{matrix} \in \mathbb{R} \right\}$$

The five vectors in the latter description form a basis.

**Problem 5**

[10 pts]

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & -2 & -3 \\ -1 & -2 & -3 & 2 & 3 \\ -1 & -2 & -3 & 2 & 3 \end{bmatrix}$$

- (a) [5 pts] Compute a basis for the column space of
- $A$
- .

All column vectors are scalar multiples of each other. So, they have the same direction. In order to span the subspace, we only need one vector.

We may take the basis to be the first column vector, i.e.,

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} \right\}$$

- (b) [5 pts] Compute the dimension of the nullspace of
- $A$
- .

$$\text{rk}(A) = \dim \text{column-space of } A = 1$$

$$\text{Therefore, nullity}(A) = \# \text{columns} - \text{rk}(A) = 5 - 1 = 4.$$

**Problem 6**

[10 pts]

- (a) [5 pts] Let
- $\mathbf{v} = (-1, 2, -3)$
- . Find
- $a \in \mathbb{R}$
- such that
- $\mathbf{v}$
- belongs to the line

$$\begin{cases} x = 4 + 10\lambda \\ y = -3 - 10\lambda \\ z = 3 + a\lambda \end{cases} \quad \lambda \in \mathbb{R}.$$

Looking at the  $x$  and  $y$  coordinate, we find that  $\lambda = -\frac{1}{2}$ . Looking at the  $z$  coordinate, we find in addition that  $a = 12$ .

- (b) [5 pts] Make the equation of the plane which is parallel to

$$-x - y + 2z = 0$$

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and which is passing through  $\mathbf{v}$ .

The equation of any plane parallel to the given differs by a scalar. So, we may assume the form

$$-x - y + 2z + d = 0$$

To determine  $d$ , we plug in the vector  $\mathbf{v}$  and find  $d = 7$ .